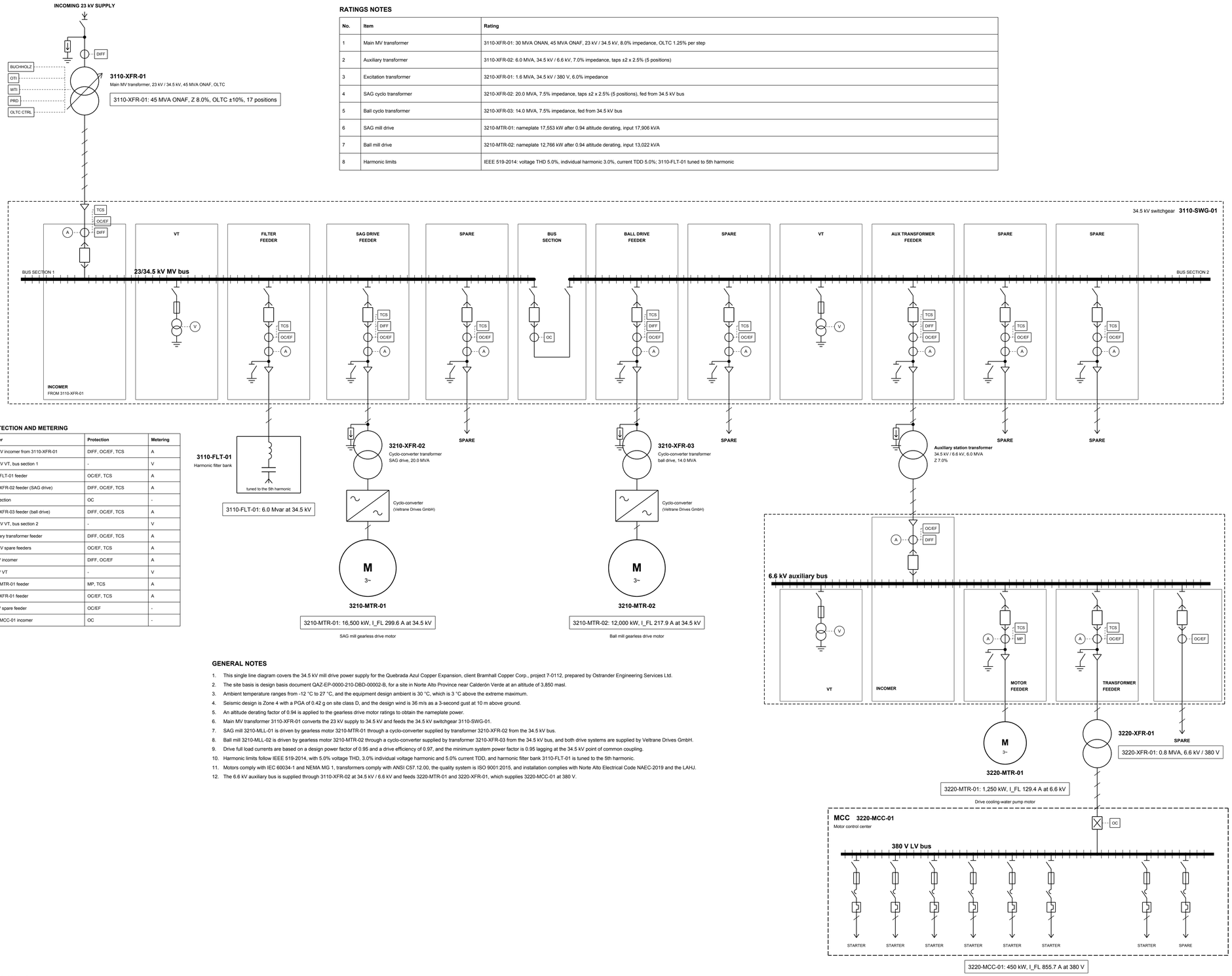


RATINGS NOTES

No.	Item	Rating
1	Main MV transformer	3110-XFR-01: 30 MVA ONAN, 45 MVA ONAF, 23 kV / 34.5 kV, 8.0% impedance, OLTC 1.25% per step
2	Auxiliary transformer	3110-XFR-02: 6.0 MVA, 34.5 kV / 6.6 kV, 7.0% impedance, taps ±2 x 2.5% (5 positions)
3	Excitation transformer	3210-XFR-01: 1.6 MVA, 34.5 kV / 380 V, 6.0% impedance
4	SAG cyclo transformer	3210-XFR-02: 20.0 MVA, 7.5% impedance, taps ±2 x 2.5% (5 positions), fed from 34.5 kV bus
5	Ball cyclo transformer	3210-XFR-03: 14.0 MVA, 7.5% impedance, fed from 34.5 kV bus
6	SAG mill drive	3210-MTR-01: nameplate 17,553 kW after 0.94 altitude derating, input 17,906 kVA
7	Ball mill drive	3210-MTR-02: nameplate 12,766 kW after 0.94 altitude derating, input 13,022 kVA
8	Harmonic limits	IEEE 519-2014: voltage THD 5.0%, individual harmonic 3.0%, current TDD 5.0%; 3110-FLT-01 tuned to 5th harmonic

LEGEND

CB withdrawable		CB fixed	
Disconnect switch		Earthing switch	
Fuse		Fuse-switch	
Contactor		Thermal overload relay	
Current transformer		Voltage transformer	
Surge arrester		OC/EF	Protection relay
Meter (A)		Two-winding transformer	
OLTC transformer		M 3~	Three-phase motor
Harmonic filter bank		Cyclo-converter	
MCC	Motor control center	Three-phase cable	
Cable termination		Earth	
Busbar		Junction	
Switchgear boundary		Protection wiring	
Outgoing feeder		OTI	Transformer accessory

PROTECTION AND METERING

Feeder	Protection	Metering
34.5 kV incomer from 3110-XFR-01	DIFF, OC/EF, TCS	A
34.5 kV VT, bus section 1	-	V
3110-FLT-01 feeder	OC/EF, TCS	A
3210-XFR-02 feeder (SAG drive)	DIFF, OC/EF, TCS	A
Bus section	OC	-
3210-XFR-03 feeder (ball drive)	DIFF, OC/EF, TCS	A
34.5 kV VT, bus section 2	-	V
Auxiliary transformer feeder	DIFF, OC/EF, TCS	A
34.5 kV spare feeders	OC/EF, TCS	A
6.6 kV incomer	DIFF, OC/EF	A
6.6 kV VT	-	V
3220-MTR-01 feeder	MP, TCS	A
3220-XFR-01 feeder	OC/EF, TCS	A
6.6 kV spare feeder	OC/EF	-
3220-MCC-01 incomer	OC	-

GENERAL NOTES

1. This single line diagram covers the 34.5 kV mill drive power supply for the Quebrada Azul Copper Expansion, client Bramhall Copper Corp., project 7-0112, prepared by Ostrander Engineering Services Ltd.
2. The site basis is design basis document QAZ-EP-0000-210-DBD-00002-B, for a site in Norte Alto Province near Calderón Verde at an altitude of 3,850 masl.
3. Ambient temperature ranges from -12 °C to 27 °C, and the equipment design ambient is 30 °C, which is 3 °C above the extreme maximum.
4. Seismic design is Zone 4 with a PGA of 0.42 g on site class D, and the design wind is 36 m/s as a 3-second gust at 10 m above ground.
5. An altitude derating factor of 0.94 is applied to the gearless drive motor ratings to obtain the nameplate power.
6. Main MV transformer 3110-XFR-01 converts the 23 kV supply to 34.5 kV and feeds the 34.5 kV switchgear 3110-SWG-01.
7. SAG mill 3210-MLL-01 is driven by gearless motor 3210-MTR-01 through a cyclo-converter supplied by transformer 3210-XFR-02 from the 34.5 kV bus.
8. Ball mill 3210-MLL-02 is driven by gearless motor 3210-MTR-02 through a cyclo-converter supplied by transformer 3210-XFR-03 from the 34.5 kV bus, and both drive systems are supplied by Veltrane Drives GmbH.
9. Drive full load currents are based on a design power factor of 0.95 and a drive efficiency of 0.97, and the minimum system power factor is 0.95 lagging at the 34.5 kV point of common coupling.
10. Harmonic limits follow IEEE 519-2014, with 5.0% voltage THD, 3.0% individual voltage harmonic and 5.0% current TDD, and harmonic filter bank 3110-FLT-01 is tuned to the 5th harmonic.
11. Motors comply with IEC 60034-1 and NEMA MG 1, transformers comply with ANSI C57.12.00, the quality system is ISO 9001:2015, and installation complies with Norte Alto Electrical Code NAEC-2019 and the LAHJ.
12. The 6.6 kV auxiliary bus is supplied through 3110-XFR-02 at 34.5 kV / 6.6 kV and feeds 3220-MTR-01 and 3220-XFR-01, which supplies 3220-MCC-01 at 380 V.

DEVICE SCHEDULE

Tag	Area	Description	Connected to	Also in documents
3110-SWG-01	3110 MV substation	34.5 kV switchgear	Fed from 3110-XFR-01; supplies drive, filter and auxiliary feeders	-
3110-XFR-01	3110 MV substation	Main MV transformer, 23 kV / 34.5 kV, 45 MVA ONAF, OLTC	23 kV supply to 3110-SWG-01 34.5 kV bus	QAZ-EP-0000-265-SPC-00006-Q, QAZ-EP-0000-265-DST-00006-Q
3110-FLT-01	3110 MV substation	Harmonic filter bank, 6.0 Mvar, 34.5 kV	34.5 kV bus of 3110-SWG-01	QAZ-EP-0000-265-SPC-00006-Q, QAZ-EP-0000-265-DST-00006-Q
3210-XFR-02	3210 drive area	Cyclo-converter transformer, SAG drive, 20.0 MVA	3110-SWG-01 34.5 kV bus to SAG mill cyclo-converter	QAZ-EP-0000-265-SPC-00006-Q, QAZ-EP-0000-265-DST-00006-Q
3210-MTR-01	3210 drive area	SAG mill gearless drive motor, 16,500 kW, 72 poles	Cyclo-converter fed by 3210-XFR-02; drives SAG mill 3210-MLL-01	QAZ-EP-0000-265-SPC-00006-Q, QAZ-EP-0000-265-DST-00006-Q, QAZ-EP-3100-255-DST-55017-Q
3210-XFR-03	3210 drive area	Cyclo-converter transformer, ball drive, 14.0 MVA	3110-SWG-01 34.5 kV bus to ball mill cyclo-converter	QAZ-EP-0000-265-SPC-00006-Q, QAZ-EP-0000-265-DST-00006-Q
3210-MTR-02	3210 drive area	Ball mill gearless drive motor, 12,000 kW, 64 poles	Cyclo-converter fed by 3210-XFR-03; drives ball mill 3210-MLL-02	QAZ-EP-0000-265-SPC-00006-Q, QAZ-EP-0000-265-DST-00006-Q, QAZ-EP-3100-255-DST-55017-Q
3220-MTR-01	3220 auxiliary area	Drive cooling-water pump motor, 1,250 kW, 6.6 kV	6.6 kV auxiliary bus fed via 3110-XFR-02	QAZ-EP-0000-265-DST-00006-Q
3220-XFR-01	3220 auxiliary area	Auxiliary transformer, 0.8 MVA, 6.6 kV / 380 V	6.6 kV auxiliary bus to 3220-MCC-01 at 380 V	QAZ-EP-0000-265-DST-00006-Q
3220-MCC-01	3220 auxiliary area	Motor control center, 450 kW load, 380 V	380 V secondary of 3220-XFR-01	QAZ-EP-0000-265-SPC-00006-Q, QAZ-EP-0000-265-DST-00006-Q

REVISIONS

Rev	Date	Description	By	Chk	App
A	2019-09-06	Preliminary, for information	HB	DK	SV
B	2019-10-02	Issued for review	HB	DK	SV

CLIENT: Bramhall Copper Corp.	CONTRACTOR: Ostrander Engineering Services Ltd.
PROJECT: Quebrada Azul Copper Expansion	PROJECT NUMBER: 7-0112

TITLE

SINGLE LINE DIAGRAM - MILL DRIVE POWER SUPPLY

DOCUMENT NUMBER	REV	DATE
QAZ-5200-265-E-001_B	B	2019-10-02

STATUS: **Issued for review** — **SHEET 1 OF 1**

DISCIPLINE: 5200 / 265 — SHEET SIZE: A1

ORIGINATOR	CHECKER	APPROVER	DOCUMENT CONTROLLER
Hollis Brandt	Dario Kessling	Simon Veil	Paulo Maren
Signature	Signature	Signature	Signature